

CHARACTERS OF THE SYMPLECTIC GROUP AT THE POWERS OF KOSTANT'S ELEMENT

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ABSTRACT. We determine the characters of $\mathrm{Sp}(2m)$ at every power of Kostant's element a_P , at which every character takes a value in $\{0, \pm 1\}$. A character vanishes exactly when $\lambda + \rho$ lies on more walls of the modular root arrangement than ρ . Otherwise its value is a signed ratio of root products, and its absolute value is a product of dimensions of representations of general linear, symplectic and odd orthogonal groups, times 1 or 2. For $\gcd(k, 2m+2)$ odd this specialises known factorisations of universal characters; the even case, beyond involutions, is new. We also list the points of the same subgroup where this closed form holds for every weight, and show that where all characters are integers the remaining factor lies in $\{1, 2, 3, 4, 6\}$.

1. INTRODUCTION

Kostant [9] singled out two elements a_Q and a_P of a compact simple group at which every irreducible character takes a value in $\{0, \pm 1\}$. At a_P the non-vanishing characters and their signs are known [10, Thm. 2.4], [5, Cor. 3.2, Prop. 3.5]. Along the line of the Coxeter element, Nadimpalli, Pattanayak and Prasad determined vanishing and, under a conjugacy hypothesis, values, and asked how both are governed at a general torsion element [18, Question 8.1]. Reeder [22, §2] showed that at any element of finite order a character is, on each congruence class of $\lambda + \rho$ as singular as the element, a constant times a product of coroot values, and computed the constant when the class is Weyl-conjugate to that of ρ . At the powers a_P^k with $\gcd(k, 2m+2) > 1$, ρ lies on affine walls and the alcove argument of [5] no longer applies; when $\gcd(k, 2m+2)$ is even, some non-vanishing characters lie on classes not conjugate to that of ρ , where [22, Prop. 2.4] does not evaluate the constant. We determine the characters of $\mathrm{Sp}(2m)$ at all powers of a_P , including those classes; the new ingredient is an identity of residue multisets (Proposition 6.1).

Setting. Let $\rho = (m, m-1, \dots, 1)$ be the half-sum of the positive roots of C_m in the standard coordinates, with the form normalised so that the short roots $\alpha = e_i \pm e_j$ have $(\alpha, \alpha) = 2$. For $\theta \in \mathbb{R}$ let $g_\theta \in \mathrm{Sp}(2m)$ be the element of the maximal torus with eigenvalues $e^{\pm 2\pi i \theta j}$, $j = 1, \dots, m$; then a weight μ takes the value $e^{2\pi i \theta(\mu, \rho)}$ on g_θ . Put $t = 2m+2$. Kostant's element is $a_P = g_{1/t}$, and for $k \geq 1$

$$a_P^k = g_{p/q}, \quad d = \gcd(k, t), \quad q = t/d, \quad p = k/d.$$

For a dominant weight λ write $\ell = \lambda + \rho$, and for $x \in \mathbb{Z}^m$

$$N_q(x) = \#\{\alpha > 0 : q \mid (x, \alpha)\}, \quad Z_q(x) = \prod_{\substack{\alpha > 0 \\ q \mid (x, \alpha)}} (x, \alpha),$$

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$$E_{p,q}(\lambda) = \sum_{\alpha > 0} \left(\left\lfloor \frac{p(\ell, \alpha)}{q} \right\rfloor - \left\lfloor \frac{p(\rho, \alpha)}{q} \right\rfloor \right),$$

the sums and products running over the positive roots $e_i \pm e_j$ ($i < j$) and $2e_i$. The residues modulo q are grouped into *folded classes* $\{c, -c\}$; the class is *fixed* if $2c \equiv 0 \pmod{q}$, that is, $c \equiv 0$ or, for q even, $c \equiv q/2$.

Theorem A. *Let λ be a dominant weight of $\mathrm{Sp}(2m)$, $k \geq 1$, and d, q, p as above.*

- (1) $N_q(\ell) \geq N_q(\rho)$, and $\chi_\lambda(a_P^k) \neq 0$ if and only if $N_q(\ell) = N_q(\rho)$. Equivalently, each folded class modulo q contains at most d of the entries $\ell_1, \dots, \ell_m$, and each fixed class at most $\lfloor d/2 \rfloor$.
- (2) If $\chi_\lambda(a_P^k) \neq 0$, then

$$\chi_\lambda(a_P^k) = (-1)^{E_{p,q}(\lambda)} \frac{Z_q(\lambda + \rho)}{Z_q(\rho)}.$$

Their absolute values have the following form. For a non-fixed class c with $0 < c < q/2$ holding n of the ℓ_i , choose signs ε_i with $\varepsilon_i \ell_i \equiv c \pmod{q}$, list the integers $y_i = (\varepsilon_i \ell_i - c)/q$ in decreasing order, and put $\nu_i^{(c)} = y_i - y_n - (n - i)$. For the class 0, with entries $\ell_1 > \dots > \ell_n$, put $\nu_i^{(0)} = \ell_i/q - (n - i + 1)$; for the class $q/2$, $\nu_i^{(q/2)} = \ell_i/q - (n - i + \frac{1}{2})$. These are partitions, and an empty class contributes the empty partition.

Theorem B. *Let λ be as in Theorem A with $\chi_\lambda(a_P^k) \neq 0$. Then*

$$|\chi_\lambda(a_P^k)| = \kappa(\lambda) \dim V_{\nu^{(0)}}^{\mathrm{Sp}_{2n_0}} \dim V_{\nu^{(q/2)}}^{\mathrm{SO}_{2n_{q/2}+1}} \prod_{0 < c < q/2} \dim V_{\nu^{(c)}}^{\mathrm{GL}_{n_c}},$$

where n_c is the number of ℓ_i in the class c , the factor for $q/2$ is omitted when q is odd, and $\kappa(\lambda) = 1$ if d is odd or if the class 0 holds fewer than $d/2$ of the ℓ_i , and $\kappa(\lambda) = 2$ otherwise.

Relation to previous work. For $\gcd(k, 2m+2) = 1$ Theorem A is the type- C case of [5, Cor. 3.2, Prop. 3.5]. For d odd, Theorems A and B follow from the identity $sp_\lambda[W] = o_\lambda[W + \{1, -1\}]$ of Ayyer and Kumari [2] together with the factorisations of Albion [3, 4], and Theorem A(2) adds the sign in closed form; at $q = 2$ they are a factorisation of the kind of Karmakar's [8, Thm. 5.1]. For d even and $q \geq 3$, the values on the classes not conjugate to that of ρ , together with the factor κ , are new. Section 2 compares the statements with [1, 2, 3, 4, 8, 18, 22] in detail.

Every rational point. The criterion holds at every rational point $g_{p/q}$, in every type (Proposition 4.1), and there a non-vanishing character of $\mathrm{Sp}(2m)$ is $\pm\kappa$ times the product Δ_λ of dimensions of Theorem B times the absolute value of a character of $\mathrm{Sp}(2s)$ or $\mathrm{SO}(2s)$, $s \leq m$, at a regular element (Theorem 11.1); Section 2 lists the cases already known. Moody and Patera computed character values at elements of finite order algorithmically, with tables in small rank [16]; Pianzola studied their character fields [19]. The powers of a_P are one of four families of points at which every character of $\mathrm{Sp}(2m)$ is an integer (Proposition 9.1); its key step is a lemma of Gannon and Schopieray [6, Lemma 4.6], and two of the four families are rational by [7, Lemma 2]. On the same line, the closed form of Theorem A(2) singles out the following points.

Theorem C. *Let $m \geq 2$ and $q \geq 2$. The identity*

$$\chi_\lambda(g_{p/q}) = (-1)^{E_{p,q}(\lambda)} \frac{Z_q(\lambda + \rho)}{Z_q(\rho)}$$

holds for every dominant weight λ with $N_q(\lambda + \rho) = N_q(\rho)$ and every p prime to q if and only if $q \mid m$, $q \mid 2m + 1$, $q \mid 2m + 2$, or $q = 6$ and $m \equiv 1 \pmod{6}$. Otherwise there is such a λ for which it fails at $p = 1$ even up to sign.

Theorem D. *Let $m \geq 2$, let $q \geq 2$ be one of the denominators of Proposition 9.1, and let λ be dominant with $\chi_\lambda(g_{p/q}) \neq 0$. Then*

$$|\chi_\lambda(g_{p/q})| = c_\lambda \Delta_\lambda, \quad c_\lambda \in \{1, 2, 3, 4, 6\}.$$

More precisely, $c_\lambda = 1$ if $q \mid m$, $q \mid 2m + 1$, or $q = 6$ and $m \equiv 1 \pmod{6}$; $c_\lambda = \kappa(\lambda) \in \{1, 2\}$ if $q \mid 2m + 2$; $c_\lambda \in \{1, 2\}$ if $q \geq 4$ is even, $q \mid 2m$ and $q \nmid m$; and $c_\lambda \in \{1, 2, 3, 4, 6\}$ if $q = 6$ and $m \equiv 4 \pmod{6}$. In the last two cases every listed value occurs.

Theorem B is the case $q \mid 2m + 2$ of Theorem D. The absolute values for $q \mid m$, $q \mid 2m + 1$, and $q \mid 2m + 2$ with d odd are known (Section 2); the list of points in Theorem C, the exclusion of the remaining points, the factor κ for d even and $q \geq 3$, and the constants for $q \geq 4$ even with $q \mid 2m$, $q \nmid m$, and for $q = 6$ with $m \equiv 4 \pmod{6}$, are new.

Section 2 compares with previous work; Sections 3–8 prove Theorems A and B, Sections 9–12 the results at every rational point and Theorems C and D; Section 13 asks two questions, and Section 14 lists the computational checks.

2. RELATION TO PREVIOUS WORK

On the combinatorial side, classical characters at a full multiset of roots of unity factor into smaller classical characters [17, 1, 3, 12, 4], and at the elements of order 2 of $\mathrm{Sp}(2n)$ Karmakar [8, Thm. 5.1] expresses the values through dimensions: as $\pm$ a power of 2 times a product of two dimensions in two extreme cases, determined by how many entries of $\lambda + \rho$ are even, and as an alternating sum of products of dimensions in the intermediate ones. In $\mathrm{Sp}(2m)$ the spectrum of every power of a_P is a full multiset of roots of unity short of two letters (Proposition 3.1).

When $\gcd(k, 2m + 2) = 1$ the element a_P^k has the spectrum of a_P (Proposition 3.1) and is conjugate to it, both products are empty for every surviving weight, and Theorem A is the type- C case of [5, Cor. 3.2, Prop. 3.5]. When $\gcd(k, 2m + 2) > 1$ the values are unbounded: Theorem B below then has a factor GL_n with $n \geq 2$, or Sp_{2n} or SO_{2n+1} with $n \geq 1$, whose highest weight can be any partition with fewer than n parts for GL_n and at most n parts for the other two, as one sees by moving the entries of ρ in that class by multiples of q . For instance, $\lambda = jq\rho$ with $j \geq 1$ gives $\ell = (jq + 1)\rho$, which lies on the same walls as ρ , and $\chi_{jq\rho}(a_P^k) = (jq + 1)^{r_q}$ with $r_q = N_q(\rho) > 0$ (for the growth of characters at elements of finite order in general, see [11]).

At $k = 1$, and so at every k prime to $2m + 2$, Theorem A is due to Cellini, Möseneder Frajria and Papi, in every type [5, Cor. 3.2, Prop. 3.5], on the basis of [10, Thm. 2.4].

For odd $d = \gcd(k, 2m + 2)$, Theorems A and B are not new in substance. Adjoining the letters 1 and -1 turns a symplectic universal character into an orthogonal one, $sp_\lambda[W] = o_\lambda[W + \{1, -1\}]$ [2, (2.13) and Lemma 3.2] (see also [15, Lemma 8.9]). For d odd, by Proposition 3.1, this turns the spectrum of a_P^k into the full alphabet $d \cdot \mu_q$, and the vanishing and the factorisation are those of Albion [3, Thm. 3.2], [4, Thm. 4.1], evaluated at d letters equal to 1; Theorem A(2) adds the sign in the closed form $(-1)^{E_{p,q}(\lambda)}$. For d even the two deleted letters are both equal to 1, and neither identity, adjoining $\{1, -1\}$ or $\{1\}$, restores a full alphabet. For d even and $q \geq 3$, the values on the classes not conjugate to that of ρ , together with the factor κ , are new.

At the other powers the results of Reeder [22, §2] apply to $t = a_P^k$, whose adjoint action has order q (for $m \geq 2$; in $\mathrm{Sp}(2)$, $a_P^2 = -1$ is central). His statistic counts the coroots $\alpha^\vee$ with $q \mid \langle \ell, \alpha^\vee \rangle$, ours the roots α with $q \mid \langle \ell, \alpha \rangle$; the two differ only on the long roots, where $\langle \ell, 2e_i \rangle = 2\ell_i$ and $\langle \ell, e_i \rangle = \ell_i$. For q odd they agree, and [22, Prop. 2.2] gives the vanishing half of Theorem A(1) and the form of Theorem A(2) up to a constant that depends on the class of ℓ modulo q ; [22, Prop. 2.4] computes that constant when the class is Weyl-conjugate to the class of ρ (his classes are cosets of qQ , finer than classes modulo q). What Theorem A adds for q odd is that the constant never vanishes on the surviving classes and its value on the classes not conjugate to that of ρ , which occur for d even (Example 7.1). For q even the hypothesis of [22, Prop. 2.2(b)] fails already at $\ell = \rho$: in $\mathrm{Sp}(6)$ at a_P^2 , where $q = 4$, the roots with $q \mid \langle \rho, \alpha \rangle$ are $e_1 + e_3$ and $2e_2$, and the only coroot with $q \mid \langle \rho, \alpha^\vee \rangle$ is $e_1 + e_3$. It still applies to part of the weights for q even, and its part (a) gives the vanishing of some others; Theorem A treats all of them at once.

Karmakar [8, Thm. 5.1] treats every class of involutions of $\mathrm{Sp}(2n)$, up to multiplication by -1 . Up to the central element -1 , the power $a_P^{m+1} = g_{1/2}$ is the balanced involution, whose eigenvalues 1 and -1 have multiplicities $2\lfloor m/2 \rfloor$ and $2\lceil m/2 \rceil$; for it the intermediate cases of [8, Thm. 5.1] do not occur, and Theorems A and B at $q = 2$ are a factorisation of the same kind as his. Theorems A and B cover every power of a_P , of every order, but among the involutions only this class, while [8] covers all of them. At $q \geq 3$ the non-fixed classes contribute factors of type GL.

On the line of $\rho^\vee$, with the coroot statistic, [18, Cor. 3.7] is the count criterion, and [18, Thm. 4.1] gives the value $\pm d_\lambda/d_m$ under the hypothesis $w(\lambda + \rho) = \rho + m\mu$. Part (1) of Theorem A is the same count on the line of ρ , with the root statistic, and is proved by the same argument (Proposition 4.1); in type C the two lines are different, the eigenvalues of $\rho^\vee(e^{2\pi i\theta})$ being $e^{\pm\pi i\theta(2j-1)}$. Part (2) needs no conjugacy hypothesis: for d even and $q \geq 3$ some non-vanishing characters have no Weyl group element carrying ℓ to ρ modulo q (Example 7.1), and the step that conjugacy provides in [18] is replaced by an identity of residue multisets (Proposition 6.1).

Theorem B has the shape of the factorisations of Ayyer and Kumari [1, Thm. 2.11] at the full alphabet $x \cdot \mu_t$, indexed by the t -quotient, and the three types on the folded classes appear in Lecouvey's plethysm [13, §3.2.2]. Here the alphabet is $d \cdot \mu_q$, the alphabet of [21, Thm. 2], with two letters deleted; the deletion changes the occupancies of the fixed classes and produces κ : when d is even exactly one class holds one entry fewer than its bound in Theorem A(1), and κ records whether it is the class 0.

At the other rational points $g_{p/q}$ (Theorem 11.1), write $m = uq + v$ with $0 \leq v < q$ and $a = \lfloor (q-1)/2 \rfloor$; several cases are known. For $q > 2m$ no (ρ, α) is divisible by q , and the statement is the classical anti-invariance of characters under the affine Weyl group, with $\Delta_\lambda = 1$. Beyond that, $v = 0$ is [1, Thm. 2.11]; $v = 1$ with $q \geq 3$ is related to [12, Thm. 2.8(3)]; $v = a$ with q odd, that is $q \mid 2m + 1$, follows from [4, Thm. 1.4] through $sp_\lambda[W] = so_\lambda^-[W + \{1\}]$; $v = a$ with q even is the case d odd of $q \mid 2m + 2$ above; and [2, Cor. 5.5] treats the weights equal to their own q -core with at most v parts. For $q \leq 2m$ and $2 \leq v \leq q-1$, $v \neq a$, the reduction is new, and so is its residual factor of type D .

3. THE ELEMENT AND ITS SPECTRUM

Let G be simple and simply connected with maximal torus T , positive roots Φ^+ and half-sum ρ , with the invariant form normalised so that short roots have $(\alpha, \alpha) = 2$. Write $\rho^\sharp \in \mathfrak{t}$ for the

vector the form identifies with ρ , and for real θ put

$$(1) \quad g_\theta = \exp(2\pi i \theta \rho^\#) \in T,$$

so that a weight μ takes the value $e^{2\pi i \theta(\mu, \rho)}$ on g_θ . Throughout, $\theta = p/q$ in lowest terms. The integer q is the denominator of the parameter and need not be the order of g_θ in G ; every statement below is about q . In $\mathrm{Sp}(2m)$ the element g_θ has eigenvalues $e^{\pm 2\pi i \theta j}$, and with t, d, q, p as in the introduction, $a_P^k = g_{1/t}^k$ has eigenvalues $\xi_q^{\pm j}$, $j = 1, \dots, m$, where $\xi_q = e^{2\pi i p/q}$ is a primitive q -th root of unity.

Proposition 3.1. *Let $t = 2m + 2$, $d = \gcd(k, t)$ and $q = t/d$. As a multiset, the spectrum $A = \{\zeta^{\pm jk} : j = 1, \dots, m\}$ of a_P^k on $\mathbb{C}^{2m}$, with ζ of order t , satisfies*

$$A = d \cdot \mu_q - 2 \cdot \{1\} \quad (d \text{ even}), \quad A = d \cdot \mu_q - \{1\} - \{-1\} \quad (d \text{ odd}).$$

Proof. Since $t = 2m + 2$, the exponents $\{\pm j : j = 1, \dots, m\}$ are exactly $\mathbb{Z}/t \setminus \{0, t/2\}$. The map $e \mapsto \zeta^{ke}$ factors through $\mathbb{Z}/q$, because ζ^k has order q , and every residue mod q is the image of exactly d elements of $\mathbb{Z}/t$; hence $\{\zeta^{ke} : e \in \mathbb{Z}/t\} = d \cdot \mu_q$. It remains to delete the two excluded exponents. For $e = 0$ the element is 1. For $e = t/2 = dq/2$: if d is even this is $\equiv 0 \pmod{q}$ and the element is again 1; if d is odd then q is even, since $dq = t$ is, and $dq/2 \equiv q/2$, so the element is -1 . $\square$

In particular the spectrum depends on d and q only, so $\chi_\lambda(a_P^k) = \chi_\lambda(a_P^d)$ and one may take $p = 1$ throughout. The full multiset $d \cdot \mu_q$ is the alphabet of [21, Thm. 2] (with m there equal to d here and n there equal to q here), and the two deleted letters are fixed points of $w \mapsto w^{-1}$.

4. THE PRODUCT FORMULA AND THE CRITERION

On the line of ρ the Weyl character formula takes the form

$$(2) \quad \chi_\lambda(g_\theta) = e^{-2\pi i \theta(\lambda, \rho)} \prod_{\alpha > 0} \frac{1 - e^{2\pi i \theta(\lambda + \rho, \alpha)}}{1 - e^{2\pi i \theta(\rho, \alpha)}}$$

for every θ at which the denominator does not vanish. Indeed, with $\ell = \lambda + \rho$ and $z = e^{2\pi i \theta}$, one has $\sum_w \epsilon(w) z^{(w\ell, \rho)} = \sum_w \epsilon(w) z^{(w\rho, \ell)}$, since $(w\ell, \rho) = (\ell, w^{-1}\rho)$ and $\epsilon(w) = \epsilon(w^{-1})$; the Weyl denominator identity turns the right-hand side into $z^{(\rho, \ell)} \prod_{\alpha > 0} (1 - z^{-(\ell, \alpha)})$, and dividing by the same computation at $\ell = \rho$ gives (2). This is the principal specialisation, written with the root statistic (ℓ, α) instead of the coroot statistic $\langle \ell, \alpha^\vee \rangle$. Since $\sum_{\alpha > 0} (\lambda, \alpha) = 2(\lambda, \rho)$, the phases cancel and

$$(3) \quad \chi_\lambda(g_\theta) = \prod_{\alpha > 0} \frac{\sin(\pi \theta(\ell, \alpha))}{\sin(\pi \theta(\rho, \alpha))}.$$

Proposition 4.1. *Let G be simple and simply connected, λ dominant, $\ell = \lambda + \rho$, $\theta = p/q$ in lowest terms, and $N_q(x) = \#\{\alpha > 0 : q \mid (x, \alpha)\}$, $r_q = N_q(\rho)$. Then $N_q(\ell) \geq r_q$, and*

$$(4) \quad \chi_\lambda(g_{p/q}) \neq 0 \iff N_q(\ell) = r_q.$$

Proof. Each factor $1 - z^j$ of (2) has a simple zero at a primitive q -th root of unity exactly when $q \mid j$, so at $z = e^{2\pi i p/q}$ the numerator vanishes to order $N_q(\ell)$ and the denominator to order r_q . A character has no poles, so the first order is at least the second; and the limit is non-zero exactly when they agree. $\square$

At $k = 1$ in type C , and more generally whenever ρ is regular, $r_q = 0$ and (4) says that no (ℓ, α) is divisible by q ; this is the condition of [5, Cor. 3.2]. On the line of $\rho^\vee$ the same count is [18, Cor. 3.7].

5. CAPACITIES IN TYPE C

In C_m the positive roots are $e_i - e_j$, $e_i + e_j$ ($i < j$) and $2e_i$. Grouping the ℓ_i into folded classes modulo q , of occupancies n_c ,

$$(5) \quad N_q(\ell) = \sum_{c \text{ not fixed}} \binom{n_c}{2} + \sum_{f \text{ fixed}} n_f^2,$$

because in a non-fixed class each pair of indices contributes exactly one of $e_i \pm e_j$, while in a fixed class it contributes both, and each index also contributes its $2e_i$: $2\binom{n}{2} + n = n^2$; two indices in different classes contribute nothing.

Corollary 5.1. *Let $q \mid 2m + 2$ and $d = (2m + 2)/q$. Then $N_q(\ell) = r_q$ if and only if each folded class modulo q contains at most d of the ℓ_i , and each fixed class at most $\lfloor d/2 \rfloor$.*

Proof. By (5) the marginal cost of one more entry is $0, 1, 2, \dots$ in a non-fixed class and $1, 3, 5, \dots$ in a fixed one, so the places costing less than d number exactly d per non-fixed class and $\lfloor d/2 \rfloor$ per fixed one. There are $\lfloor q/2 \rfloor + 1$ folded classes, of which one is fixed when q is odd and two when q is even; since $m = dq/2 - 1$, the places of cost less than d number exactly m if d is odd and $m + 1$ if d is even. By Proposition 4.1 the minimum of N_q over the vectors $\ell = \lambda + \rho$ is r_q , and it is attained at ρ . It is also the minimum over all of $\mathbb{Z}^m$, hence over all placements of m entries in the folded classes: every $x \in \mathbb{Z}^m$ is congruent modulo q to a vector $\lambda + \rho$ with λ dominant, namely $x_i + (m - i + 1)Mq$ for a large integer M , whose entries are positive and strictly decreasing. If d is odd, reaching the minimum forces occupying all m cheap places, and any overflow replaces a place of cost $< d$ by one of cost $\geq d$. If d is even, the last admissible place of every class costs $d - 1$, so omitting any one of them gives the same minimum, while exceeding a capacity again costs strictly more. So $N_q(\ell) = r_q$ if and only if no capacity is exceeded. $\square$

When d is odd the capacities are saturated: every surviving ℓ has the occupancies of ρ . When d is even exactly one class falls one entry below its capacity; for $\rho = (m, \dots, 1)$ it is the class 0, since $m = dq/2 - 1$ and the multiples of q in $\{1, \dots, m\}$ are $q, \dots, (d/2 - 1)q$.

6. THE RESIDUE LEMMA

For the value one needs more than the number of vanishing factors in (3): the non-vanishing factors of numerator and denominator must have the same product of absolute values. This follows from an identity of residue multisets.

Proposition 6.1. *Work in $\mathbb{Z}[X]/(X^q - 1)$. For $x = (x_1, \dots, x_m) \in \mathbb{Z}^m$ put*

$$P_x(X) = \sum_{i=1}^m (X^{x_i} + X^{-x_i}), \quad \mathcal{R}_x(X) = \sum_{\alpha \in \Phi(C_m)} X^{(x, \alpha)}.$$

Then

$$(6) \quad \mathcal{R}_x(X) = \frac{P_x(X)^2 + P_x(X^2)}{2} - m,$$

and, for $q \mid 2m + 2$ and every λ satisfying the capacities of Corollary 5.1, $\mathcal{R}_{\lambda+\rho} = \mathcal{R}_\rho$.

Proof. Expanding $P_x(X)^2$ collects $X^{\pm x_i \pm x_j}$ over all ordered pairs. The off-diagonal terms give each root $\pm e_i \pm e_j$ twice; the diagonal terms give $X^{\pm 2x_i}$ once each and $2m$ copies of X^0 ; and $P_x(X^2)$ supplies the second copy of each $X^{\pm 2x_i}$. Halving therefore leaves every root of C_m exactly once, together with m copies of X^0 , and subtracting m removes those. That is (6).

For the second claim write $T_q(X) = 1 + X + \cdots + X^{q-1}$, so that $T_q(X) X^c = T_q(X)$ for every c . If d is odd the capacities are saturated, the folded occupancies of ℓ are those of ρ , and $P_\ell = P_\rho$ outright. If d is even, one class c falls a single unit below its capacity, and

$$P_\ell(X) = dT_q(X) - X^c - X^{-c}, \quad P_\rho(X) = dT_q(X) - 2,$$

so that $P_\ell(X)^2 - P_\rho(X)^2 = X^{2c} + X^{-2c} - 2$ while $P_\ell(X^2) - P_\rho(X^2) = -X^{2c} - X^{-2c} + 2$. The two corrections cancel in (6), and $\mathcal{R}_\ell = \mathcal{R}_\rho$. $\square$

Since Φ contains each root together with its negative, $\mathcal{R}_\ell = \mathcal{R}_\rho$ says that the two multisets

$$(7) \quad \{ \pm(\ell, \alpha) \bmod q : \alpha > 0 \} \quad \text{and} \quad \{ \pm(\rho, \alpha) \bmod q : \alpha > 0 \}$$

of residues taken up to sign coincide; in particular their non-zero parts coincide.

Remark 6.2. Identity (6) is the type- C instance of $\mathcal{R}_x = \mathrm{ch} \mathfrak{g}(x) - \mathrm{rank}$, valid for every simple $\mathfrak{g}$, since the weights of the adjoint representation are the roots and rank copies of 0; in type C the adjoint representation is Sym^2 of the natural one.

7. THE VALUE: PROOF OF THEOREM A

Proof of Theorem A. Part (1) is Proposition 4.1 together with Corollary 5.1, since $a_P^k = g_{p/q}$.

For part (2), let $\theta = p/q + \varepsilon$ with $\varepsilon \rightarrow 0$ in (3); the character is continuous in θ . For a positive integer A with $q \nmid A$ one has $\mathrm{sgn} \sin(\pi p A / q) = (-1)^{\lfloor p A / q \rfloor}$, and for $q \mid A$, $\sin(\pi A \theta) = (-1)^{p A / q} \pi A \varepsilon + O(\varepsilon^3)$. For λ dominant every (ℓ, α) and every (ρ, α) with $\alpha > 0$ is a positive integer. By part (1) the numerator and the denominator of (3) carry the same number r_q of vanishing factors, so the powers of $\pi \varepsilon$ cancel, each vanishing factor leaves its exponent, and every factor leaves its sign. Hence

$$\chi_\lambda(a_P^k) = (-1)^{E_{p,q}(\lambda)} \frac{Z_q(\ell)}{Z_q(\rho)} \cdot \frac{\prod_{\alpha > 0, q \nmid (\ell, \alpha)} |\sin(\pi p(\ell, \alpha)/q)|}{\prod_{\alpha > 0, q \nmid (\rho, \alpha)} |\sin(\pi p(\rho, \alpha)/q)|}.$$

The quantity $|\sin(\pi p j / q)|$ depends only on j modulo q and up to sign, so by Proposition 6.1 and (7) the last quotient is 1. $\square$

Example 7.1. In $\mathrm{Sp}(4)$ at a_P^2 one has $t = 6$, $d = 2$, $q = 3$. For $\lambda = (1, 0)$, $\ell = (3, 1)$ and $\rho = (2, 1)$; the root values are 2, 4, 6, 2 and 1, 3, 4, 2 (in the order $e_1 - e_2$, $e_1 + e_2$, $2e_1$, $2e_2$), so $N_3(\ell) = N_3(\rho) = 1$, $Z_3(\ell)/Z_3(\rho) = 6/3$, $E_{1,3}(\lambda) = 3 - 2$, and $\chi_{(1,0)}(a_P^2) = -2$, as the eigenvalues $\omega^{\pm 1}, \omega^{\pm 2}$ ($\omega = e^{2\pi i/3}$) confirm. Modulo 3 the signed permutations of ℓ are $(0, \pm 1)$ and $(\pm 1, 0)$, none of which is $\rho \equiv (2, 1)$: there is no w in the Weyl group with $w\ell \equiv \rho \pmod{3}$. The weights $\lambda = (1, 0)$ and $\lambda = (1, 1)$ occupy the same folded classes and have values -2 and $+2$ (for $\lambda = (1, 1)$, $\ell = (3, 2)$ and the root values are 1, 5, 6, 4), so the sign is not a function of the occupancies.

Remark 7.2. The sign needs only the residues of ℓ , in order. Writing $x_i = qh_i + r_i$ with $0 \leq r_i < q$, the quotients contribute $2 \sum_i (m - i + 1) h_i$ to $\sum_{\alpha > 0} \lfloor (x, \alpha) / q \rfloor$, which is even, and what is left is congruent modulo 2 to

$$F_q(r_1, \dots, r_m) = \#\{i : 2r_i \geq q\} + \#\{i < j : r_i < r_j\} + \#\{i < j : r_i + r_j \geq q\},$$

so that, on the surviving weights and with $p = 1$,

$$\operatorname{sgn} \chi_\lambda(a_P^k) = (-1)^{F_q(\ell \bmod q) - F_q(\rho \bmod q)}.$$

At $k = 1$ compare [5, Rem. 3.2], where the sign is read off a signed permutation of the residues.

8. THE FACTORISATION: PROOF OF THEOREM B

Proof of Theorem B. By Theorem A, $|\chi_\lambda(a_P^k)| = Z_q(\ell)/Z_q(\rho)$. The roots α with $q \mid (\ell, \alpha)$ join two indices of the same folded class, or are $2e_i$ with ℓ_i in a fixed class. We compute Z_q class by class.

A non-fixed class c with n entries. For two of its entries exactly one of $(\ell, e_i - e_j)$, $(\ell, e_i + e_j)$ is divisible by q , and its absolute value is $|\varepsilon_i \ell_i - \varepsilon_j \ell_j| = q|y_i - y_j|$; no $2e_i$ contributes. The y_i are distinct, and with $y_1 > \dots > y_n$ the Weyl dimension formula for GL_n gives $\prod_{i < j} (y_i - y_j) = V_n \dim V_{\nu^{(c)}}^{\mathrm{GL}_n}$, where $V_n = \prod_{j < n} j!$. The contribution is $q^{\binom{n}{2}} V_n \dim V_{\nu^{(c)}}^{\mathrm{GL}_n}$.

The class 0 with n entries $\ell_i = qz_i$, $z_1 > \dots > z_n \geq 1$. All of $e_i \pm e_j$ and $2e_i$ contribute, with values $q(z_i \pm z_j)$ and $2qz_i$. Since $z = \nu^{(0)} + \rho_{C_n}$, the Weyl dimension formula for Sp_{2n} gives the contribution $q^{n^2} C_n \dim V_{\nu^{(0)}}^{\mathrm{Sp}_{2n}}$, where $C_n = 2^n \prod_{i < j} ((n-i+1)^2 - (n-j+1)^2) n! = 2^n \prod_{j \leq n} (2j-1)!$.

The class $q/2$ with n entries $\ell_i = qz_i$, $z_i \in \frac{1}{2} + \mathbb{Z}$. Again all of $e_i \pm e_j$ and $2e_i$ contribute, and since $z = \nu^{(q/2)} + \rho_{B_n}$ with $\rho_{B_n} = (n - \frac{1}{2}, \dots, \frac{1}{2})$, the Weyl dimension formula for SO_{2n+1} gives $q^{n^2} B_n \dim V_{\nu^{(q/2)}}^{\mathrm{SO}_{2n+1}}$, where $B_n = \prod_{j \leq n} (2j-1)!$.

The same computation applies to ρ , whose entries in each class form an unbroken run (the class c receives $c, q+c, 2q+c, \dots$ and $q-c, 2q-c, \dots$, so the y_i are consecutive integers; the class 0 receives $q, 2q, \dots$; the class $q/2$ receives $q/2, 3q/2, \dots$), so all its partitions are empty and all its dimensions are 1. The powers of q agree on the two sides, since $N_q(\ell) = N_q(\rho)$. Hence $Z_q(\ell)/Z_q(\rho)$ is the product of the dimensions in the statement times the ratio of the constants $V_{n_c}, C_{n_0}, B_{n_{q/2}}$ for the occupancies of ℓ and of ρ . If d is odd the occupancies agree (Section 5) and the ratio is 1. If d is even, put $h = d/2$; the class 0 of ρ holds $h-1$ entries and every other class is full; if the class of ℓ that falls short is 0, the occupancies agree again; otherwise that class holds one entry less and the class 0 one entry more than for ρ , and the ratio is

$$\frac{C_h/C_{h-1}}{V_{2h}/V_{2h-1}} = \frac{2(2h-1)!}{(2h-1)!} = 2 \quad \text{or} \quad \frac{C_h/C_{h-1}}{B_h/B_{h-1}} = 2$$

according as that class is non-fixed or is $q/2$. □

Example 8.1. In $\mathrm{Sp}(10)$ at a_P^4 (so $d = 4$, $q = 3$) take $\lambda = (3, 2, 1, 0, 0)$. Then $\ell = (8, 6, 4, 2, 1)$, $Z_3(\ell)/Z_3(\rho) = 8$ and $E_{1,3}(\lambda) = 17$, so $\chi_\lambda(a_P^4) = -8$. The class 0 holds $\ell_2 = 6$, one entry below its capacity, so $\kappa = 1$; the class $\{1, 2\}$ holds the other four, with $y = (1, 0, -1, -3)$; and the factorisation reads $\dim V_{(1,1,1,0)}^{\mathrm{GL}_4} = 4$ times $\dim V_{(1)}^{\mathrm{Sp}_2} = 2$.

When d is odd every surviving ℓ has the occupancies of ρ , so the product of groups in Theorem B is the same for every λ , and λ only selects the highest weights. At $d = 1$ every class holds at most one entry, all dimensions are 1, and Theorem B recovers $|\chi_\lambda(a_P)| = 1$. The groups are not centralisers: for $m = 6$ and $k = 7$ one has $d = 7$, $q = 2$, and the factorisation group is $\mathrm{Sp}_6 \times \mathrm{SO}_7$, of type $C_3 \times B_3$, while the centraliser of a_P^7 in $\mathrm{Sp}(12)$, whose eigenvalues are 1 and -1 with multiplicity six each, is $\mathrm{Sp}_6 \times \mathrm{Sp}_6$. No representation of the centraliser serves either: for

$\lambda = (1, 1, 0, 0, 0, 0)$ the defining representation has trace 0 at a_P^7 and its exterior square trace -6 , so $\chi_\lambda(a_P^7) = -7$, while $\mathrm{Sp}_6 \times \mathrm{Sp}_6$ has no irreducible representation of dimension 7.

9. WHICH POINTS OF THE LINE HAVE INTEGER VALUES

Proposition 4.1 holds at every rational point of the line, and it is natural to ask what distinguishes the powers of a_P among those points.

Proposition 9.1. *Let $m \geq 2$ and $\gcd(p, q) = 1$. Then $\chi_\lambda(g_{1/q}) \in \mathbb{Z}$ for every dominant weight λ of $\mathrm{Sp}(2m)$ if and only if at least one of*

$$q \mid 2m, \quad q \mid 2m + 1, \quad q \mid 2m + 2, \quad q = 6 \text{ and } m \equiv 1 \pmod{3}$$

holds; and in that case $g_{p/q}$ is conjugate to $g_{1/q}$ for every p coprime to q , so no character value depends on the numerator.

The step on fields in the proof below is [6, Lemma 4.6]: in their notation $\beta_w = [2w + 1]_q$, and their lemma gives $\mathbb{Q}(\beta_w) = \mathbb{Q}(\cos(2\pi/q))$ for $1 \leq w < a$. For $q \mid 2m + 1$ and $q \mid 2m + 2$ the sufficiency also follows from [7, Lemma 2]: the regular elements of order $h + 1 = 2m + 1$ and Kostant's element a_P of type ρ , of order $h + 2$, are conjugate to rational points of the group, so their powers have integer characters. That a torsion element has rational character values exactly when it is conjugate to its powers prime to its order is a criterion of Moody and Patera [16, Prop. 5.1] (see [20, p. 512]). On the line of $\rho^\vee$, Pianzola determined the decomposition groups of the corresponding elements for denominators at least $2h - 2$ [19, Prop. 18]; Proposition 9.1 concerns the line of ρ and every q . We include the short argument, which is elementary, for completeness.

Proof. Suppose every $\chi_\lambda(g_{1/q})$ is an integer. The exterior powers $\Lambda^j \mathbb{C}^{2m}$ are direct sums of irreducibles, so their characters are $\mathbb{Z}$ -combinations of the χ_λ and are integers as well; those characters are the coefficients of the characteristic polynomial of $g_{1/q}$ on $\mathbb{C}^{2m}$, so that polynomial lies in $\mathbb{Z}[X]$, and the multiplicities of its roots are constant along Galois orbits. Conversely, if the spectrum is carried to itself by every $\xi_q \mapsto \xi_q^p$, then $g_{p/q}$ and $g_{1/q}$ have the same spectrum and are conjugate in $\mathrm{Sp}(2m)$; the character values are then algebraic integers fixed by the Galois group of $\mathbb{Q}(\xi_q)$, hence rational integers.

Write $m = uq + v$ with $0 \leq v < q$ and $a = \lfloor (q - 1)/2 \rfloor$. The spectrum $\{\xi_q^{\pm j} : j = 1, \dots, m\}$ is, as a multiset of μ_q ,

$$2u \cdot \mu_q + \{\pm 1, \dots, \pm v\},$$

and the first summand is Galois-stable. So the question is when the segment $\{\pm 1, \dots, \pm v\}$ is. It is stable for $v = 0$, $v = a$, $v = q - 1$ and, when q is even, $v = q/2$. For necessity write $A_v = \sum_{j=1}^v (\xi_q^j + \xi_q^{-j})$; if the segment is stable then A_v is fixed by the whole Galois group. A full turn sums to zero, whence $A_v + A_{q-1-v} = -2$ and $\mathbb{Q}(A_v) = \mathbb{Q}(A_w)$ for $w = \min(v, q - 1 - v)$; the four segments just listed are $w = 0$ and $w = a$, so only $1 \leq w < a$ is left. There set $\beta_w = 1 + A_w = \sum_{j=-w}^w \xi_q^j$, whose conjugates are

$$\sigma_p(\beta_w) = \frac{\sin((2w + 1)\pi p/q)}{\sin(\pi p/q)}.$$

For $q \geq 4$, $2 \leq N \leq q - 2$ and $1 \leq r \leq q - 1$ with $r \neq 1, q - 1$, one has the strict inequality

$$\left| \frac{\sin(N\pi r/q)}{\sin(\pi r/q)} \right| < \frac{\sin(N\pi/q)}{\sin(\pi/q)}.$$

Indeed, replacing N by $n = \min(N, q - N)$ changes no modulus; on $0 < x < \pi/n$ the function $f(x) = \sin(nx)/\sin x$ is positive and strictly decreasing, since $f'/f = n \cot(nx) - \cot x < 0$ because $t \cot t$ decreases on $(0, \pi)$; and on $\pi/n \leq x \leq \pi/2$ one has $|f(x)| \leq 1/\sin(\pi/n) < 1/\sin(\pi/2n) = f(\pi/2n) \leq f(\pi/q)$. Taking $N = 2w + 1$, which lies in that range precisely when $1 \leq w < a$, the only automorphisms fixing β_w are the identity and complex conjugation; so $\mathbb{Q}(\beta_w) = \mathbb{Q}(\xi_q + \xi_q^{-1})$, which is $\mathbb{Q}$ only when $\varphi(q) = 2$. Inside this range that leaves $q = 6$, $w = 1$, that is $(q, v) = (6, 1)$ and $(6, 4)$. Translating the surviving residues into divisibilities gives the four conditions. $\square$

Remark 9.2. The restriction to rational parameters is no loss. If $\chi_\lambda(g_\theta) \in \mathbb{Z}$ for every dominant λ , for some real θ , then, as in the proof above, the characteristic polynomial of g_θ on $\mathbb{C}^{2m}$ is monic with integer coefficients. Its roots $e^{\pm 2\pi i \theta j}$ all have absolute value 1, and the conjugates of a root are again roots, so by Kronecker's theorem they are roots of unity; since $e^{2\pi i \theta}$ is one of them, θ is rational.

The powers of a_P form the family $q \mid 2m + 2$. Integrality is weaker than the closed form of Theorem A: in $\mathrm{Sp}(4)$ at $q = 4$, a point of the family $q \mid 2m$, the weight $\lambda = (1, 0)$ has $Z_4(\ell)/Z_4(\rho) = 1$ while $\chi_{(1,0)}(g_{1/4}) = -2$. Away from the four families the closed form does not hold uniformly: in $\mathrm{Sp}(4)$ at $q = 7$ the weight $\lambda = (1, 0)$ has $Z_7(\ell)/Z_7(\rho) = 1$ while $\chi_{(1,0)}(g_{1/7}) = 2 \cos \frac{2\pi}{7} + 2 \cos \frac{4\pi}{7}$, whereas the weight $\lambda = (7, 7)$, with root values 1, 17, 18, 16 against 1, 3, 4, 2 for ρ , has $\chi_{(7,7)}(g_{1/7}) = 1 = Z_7(\ell)/Z_7(\rho)$. Theorem 11.1 gives the value at every denominator.

10. HOW MANY CHARACTERS SURVIVE

For a root system X with weight lattice P write $S_X(q) = \#\{x \in P/qP : N_q(x) = r_q\}$, the number of residues of weights at which the criterion (4) admits a non-vanishing character. In type C it is given in closed form for every q . Put

$$f = \begin{cases} 1, & q \text{ odd}, \\ 2, & q \text{ even}, \end{cases} \quad a = \frac{q-f}{2}, \quad m = uq + v \quad (0 \leq v < q).$$

Theorem 10.1.

$$S_{C_m}(q) = \begin{cases} \frac{m! 2^{2ua+v} \binom{a}{v}}{((2u+1)!)^v ((2u)!)^{a-v} (u!)^f}, & v \leq a, \\ \frac{m! 2^{a(2u+1)} \binom{a+f}{v-a}}{((2u+1)!)^a (u!)^f (u+1)^{v-a}}, & v > a. \end{cases}$$

Proof. By (5) the cost of an extra entry is 0, 1, 2, ... in a non-fixed class and 1, 3, 5, ... in a fixed one, so the minimum of N_q is attained by filling the cheapest places, and $\rho = (m, \dots, 1)$ attains it, its residues being u full turns of $\mathbb{Z}/q$ followed by $1, \dots, v$; by Proposition 4.1, and since every placement of entries in classes is the residue of some $\lambda + \rho$ (proof of Corollary 5.1), that minimum is r_q . When $v \leq a$, each fixed class carries u entries, v non-fixed classes carry $2u + 1$ and the remaining $a - v$ carry $2u$; the binomial $\binom{a}{v}$ chooses the v classes, the multinomial coefficient distributes the m coordinates, and each entry of a non-fixed class has two possible residues, which gives the power of 2. When $v > a$ one starts from $2u + 1$ in every non-fixed class and u in every fixed one and places the remaining $v - a$ entries in distinct classes, chosen in $\binom{a+f}{v-a}$ ways; adding an entry multiplies the count by $1/(u+1)$ whether the class is fixed or not. $\square$

For $q > 2m$ one has $u = 0$ and $v = m \leq a$, so $S_{C_m}(q) = m! 2^m \binom{a}{m}$, which is $\prod_i (q - (2i - 1))$ for q odd and $\prod_i (q - 2i)$ for q even. For $q \mid 2m + 2$ it is the multinomial count over the capacities of Corollary 5.1, and at a_P itself $S_{C_m}(2m + 2) = 2^m m!$. For instance $S_{C_3}(5) = 72$ and $S_{C_3}(6) = 96$. The function $q \mapsto S_{C_m}(q)$ is not a quasi-polynomial: $S_{C_2}(q) = (q - 1)(q - 3)$ for every large odd q , while $S_{C_2}(3) = 8$; the characteristic quasi-polynomial of the modular arrangement counts the complement of the walls, and agrees with $S_X(q)$ only where $r_q = 0$.

11. THE VALUE AT EVERY RATIONAL POINT

Proposition 4.1 and the limit in the proof of Theorem A hold at every rational point $g_{p/q}$, but in general the residue multisets of ℓ and ρ differ, and the quotient of sines in that proof is no longer 1. It is the absolute value of a character of a symplectic or even orthogonal group of rank at most m at a regular element; the rank is m when $q > 2m$, where $Y = C_m$, and when $q = 2m$, where $Y = D_m$.

Let $q \geq 2$, $\gcd(p, q) = 1$, and $m = uq + v$, f and a as in Section 10; represent each folded class by its element c with $0 \leq c \leq q/2$. For λ with $N_q(\ell) = r_q$ the proof of Theorem 10.1 gives the occupancies: if $v \leq a$, each fixed class holds u entries, v non-fixed classes hold $2u + 1$ and the others $2u$; if $v > a$, every class holds $2u + 2$ entries if it is non-fixed and $u + 1$ if it is fixed, except $q - v$ classes, which hold one entry fewer. Indeed, the marginal costs listed in that proof increase within each class, so a placement minimises N_q exactly when it uses the places of cost below a threshold and some places of cost equal to it; the threshold is $2u$ when $v \leq a$, and $2u + 1$ when $v > a$. Let $b_1 > \dots > b_s$ be the classes with $2u + 1$ entries when $v \leq a$, so that $s = v$, and the classes with one entry fewer when $v > a$, so that $s = q - v$. Put $Y = C_s$, $\rho_Y = (s, \dots, 1)$ in the first case and $Y = D_s$, $\rho_Y = (s - 1, \dots, 0)$ in the second, let g_θ^Y be the element of the maximal torus of $\mathrm{Sp}(2s)$, respectively $\mathrm{SO}(2s)$, with eigenvalues $e^{\pm 2\pi i \theta_j}$, j running over the entries of ρ_Y , and let χ_μ^Y be the irreducible character of highest weight μ . Two degenerate cases are included: for $s = 0$, that is $Y = C_0$, the group is trivial and $\chi^Y = 1$; for $Y = D_1$ the group is $\mathrm{SO}(2)$, $\rho_Y = (0)$, g_θ^Y is the identity and $\chi_\mu^Y(g_\theta^Y) = 1$. In both cases Y has no roots and the regularity of g_θ^Y is vacuous. Let Δ_λ be the product of dimensions in Theorem B, with the partitions $\nu^{(c)}$ defined as there, and put $\kappa = 2$ if $v > a$ and the class 0 holds $u + 1$ entries, $\kappa = 1$ otherwise.

Theorem 11.1. *Let λ be a dominant weight of $\mathrm{Sp}(2m)$ with $N_q(\lambda + \rho) = r_q$. Then $b - \rho_Y$ is a dominant weight of Y , the element $g_{p/q}^Y$ is regular, and*

$$\chi_\lambda(g_{p/q}) = (-1)^{E_{p,q}(\lambda)} \kappa \Delta_\lambda |\chi_{b-\rho_Y}^Y(g_{p/q}^Y)|.$$

For $v \in \{0, a, q - 1\}$, that is, for $q \mid m$, $q \mid 2m + 1$ or $q \mid 2m + 2$, the last factor is 1 for every such λ ; for $q \mid 2m + 2$ the theorem is Theorems A(2) and B.

Proof. The class of ρ . The residues of ρ are u full turns of $\mathbb{Z}/q$ followed by $1, \dots, v$. If $v \leq a$ the classes with $2u + 1$ entries are $1, \dots, v$. If $v > a$ the classes with one entry fewer are 0 and the non-fixed $c < q - v$; the class $q/2$, for q even, is full, since $v \geq q/2$. In both cases these classes are the entries of ρ_Y .

Regularity. The entries of b are distinct and lie in $\{1, \dots, a\}$ if $Y = C_s$, and in $\{0, \dots, a\}$, together with $q/2$ when q is even, if $Y = D_s$. Hence every (b, α) with α a positive root of Y , namely $b_i \pm b_j$ and, for C_s , $2b_i$, lies strictly between 0 and q , since $2a < q$ and $a + q/2 < q$ for q even. The same holds for ρ_Y , which is one of the possible b . So $b - \rho_Y$ is dominant and $g_{p/q}^Y$ is

regular. The computation that gives (3) in Section 4 is stated there for G simple and simply connected, but it uses only the Weyl character formula and the Weyl denominator identity, with ρ a weight of the group; it therefore applies verbatim to $\mathrm{Sp}(2s)$ and to $Y = \mathrm{SO}(2s)$, where ρ_Y is a weight, including D_2 , which is not simple, and D_1 , a torus, where both sides are 1. It gives for Y

$$|\chi_{b-\rho_Y}^Y(g_{p/q}^Y)| = \frac{\Pi^Y(b)}{\Pi^Y(\rho_Y)}, \quad \Pi^Y(x) = \prod_{\alpha \in \Phi^+(Y)} |\sin(\pi p(x, \alpha)/q)|.$$

Residues. In $\mathbb{Z}[X]/(X^q - 1)$, with P_x and T_q as in Proposition 6.1, a non-fixed class c holding n entries contributes $n(X^c + X^{-c})$ to P_ℓ , and a fixed class c contributes $2nX^c = n(X^c + X^{-c})$. Hence

$$P_\ell = 2uT_q + P_b \quad (v \leq a), \quad P_\ell = 2(u+1)T_q - P_b \quad (v > a),$$

and the same holds for ρ with ρ_Y in place of b . Since $T_q^2 = qT_q$ and $T_q P_x = 2sT_q$ for $x \in \mathbb{Z}^s$, every term of (6) that contains $T_q(X)$ or $T_q(X^2)$ is the same for ℓ and for ρ , and

$$\mathcal{R}_\ell - \mathcal{R}_\rho = \frac{P_b^2 \pm P_b(X^2)}{2} - \frac{P_{\rho_Y}^2 \pm P_{\rho_Y}(X^2)}{2} = \mathcal{R}_b^Y - \mathcal{R}_{\rho_Y}^Y,$$

with the upper sign for $v \leq a$. Here $\mathcal{R}_x^Y = \sum_{\alpha \in \Phi(Y)} X^{(x, \alpha)}$, which is $(P_x^2 + P_x(X^2))/2 - s$ for C_s by (6), and $(P_x^2 - P_x(X^2))/2 - s$ for D_s , whose roots $\pm e_i \pm e_j$ are those of C_s without the $\pm 2e_i$.

The value. The limit in the proof of Theorem A uses only $N_q(\ell) = r_q$, so

$$\chi_\lambda(g_{p/q}) = (-1)^{E_{p,q}(\lambda)} \frac{Z_q(\ell)}{Z_q(\rho)} \cdot \frac{\Pi(\ell)}{\Pi(\rho)}, \quad \Pi(x) = \prod_{\alpha > 0, q \nmid (x, \alpha)} |\sin(\pi p(x, \alpha)/q)|.$$

Now $\Pi(x)^2$ is the product of $|\sin(\pi pc/q)|$ over the non-zero residues c of $\mathcal{R}_x$, counted with multiplicity, and $\Pi^Y(x)^2$ is the same product for $\mathcal{R}_x^Y$, none of whose residues is zero. The identity of residue multisets $\mathcal{R}_\ell + \mathcal{R}_{\rho_Y}^Y = \mathcal{R}_\rho + \mathcal{R}_b^Y$ therefore gives $\Pi(\ell)/\Pi(\rho) = \Pi^Y(b)/\Pi^Y(\rho_Y)$.

The dimensions. As in the proof of Theorem B, $Z_q(\ell)/Z_q(\rho)$ is Δ_λ times the ratio of the constants V_n, C_n, B_n for the occupancies of ℓ and of ρ , since the dimensions for ρ are 1 and the powers of q agree. If $v \leq a$ the two multisets of occupancies agree class type by class type and the ratio is 1. If $v > a$, the occupancies agree outside the classes that fall short, a class falling short multiplies the constant by $V_{2u+1}/V_{2u+2} = 1/(2u+1)!$, by $B_u/B_{u+1} = 1/(2u+1)!$ or, for the class 0, by $C_u/C_{u+1} = 1/(2(2u+1)!)$, and both ℓ and ρ have $q-v$ such classes, the class 0 among them for ρ . The ratio is 2 if the class 0 is full for ℓ and 1 otherwise, that is, κ .

The cases $v \in \{0, a, q-1\}$. If $v = 0$ then $s = 0$ and $Y = C_0$. If $v = a$ then $Y = C_a$, and the a distinct entries of b in $\{1, \dots, a\}$ give $b = (a, \dots, 1) = \rho_Y$. If $v = q-1$ then $Y = D_1$. In the three cases the last factor is 1. Now $v = 0$ means $q \mid m$; $v = a$ means $2m+1 = (2u+1)q$ for q odd and $2m+2 = (2u+1)q$ for q even; and $v = q-1$ means $q \mid m+1$. For $q \mid 2m+2$, with $d = (2m+2)/q$, the case d odd is $v = a$ with q even, where $\kappa = 1$, and the case d even is $v = q-1$, where $u+1 = d/2$; in both, κ is the $\kappa(\lambda)$ of Theorem B. $\square$

Corollary 11.2. *Let $q \mid m$ or $q \mid 2m+1$, let $\gcd(p, q) = 1$, and let λ be a dominant weight of $\mathrm{Sp}(2m)$. Then $\chi_\lambda(g_{p/q}) \neq 0$ if and only if each non-fixed folded class modulo q contains at most $2m/q$, respectively $(2m+1)/q$, of the entries of $\ell = \lambda + \rho$, and the fixed classes at most m/q , respectively $(2m+1-q)/(2q)$; these bounds are then attained. In that case*

$$\chi_\lambda(g_{p/q}) = (-1)^{E_{p,q}(\lambda)} \frac{Z_q(\lambda + \rho)}{Z_q(\rho)}, \quad |\chi_\lambda(g_{p/q})| = \Delta_\lambda,$$

where Δ_λ is the product of dimensions of Theorem B.

Proof. The two cases are $v = 0$, with $u = m/q$, and $v = a$ with q odd, with $u = (2m+1-q)/(2q)$. In both $v \leq a$, so the occupancies of the surviving weights, listed before Theorem 11.1, are $2u$ or $2u+1$ in every non-fixed class and u in every fixed class; they add up to m , so the bounds are attained, and a placement exceeding one of them costs more than r_q by (5) and the marginal costs in the proof of Theorem 10.1. Since $v \leq a$, $\kappa = 1$, and the last factor of Theorem 11.1 is 1. Its proof gives $\chi_\lambda(g_{p/q}) = (-1)^{E_{p,q}(\lambda)} (Z_q(\ell)/Z_q(\rho)) \Pi(\ell)/\Pi(\rho)$ with $\Pi(\ell)/\Pi(\rho)$ equal to that factor, and $Z_q(\ell)/Z_q(\rho) = \kappa \Delta_\lambda$. $\square$

The case $q \mid 2m+1$ is one of the four families of Proposition 9.1, and $q \mid m$ is part of the family $q \mid 2m$. In the rest of that family, with q even and $q \nmid m$, one has $v = q/2 > a$ and $Y = D_{q/2}$, and the last factor is not always 1: in $\mathrm{Sp}(4)$ at $q = 4$ the weight $\lambda = (1, 0)$ has $Z_4(\ell)/Z_4(\rho) = 1$ and value -2 (Section 9). Section 12 determines it.

Some cases of Theorem 11.1 are known. For $v = 0$ the spectrum of $g_{p/q}$ is the full alphabet $2u \cdot \mu_q$, and the value is the case $X = (1, \dots, 1)$ of [1, Thm. 2.11]; for $u = 1$ see also [2, Thm. 7.2]. For $v = 1$ and $q \geq 3$ the value is closely related to the factorisation [12, Thm. 2.8(3), eq. (2.17)] at $X = (1, \dots, 1)$ and $y = \xi_q$, a product of dimensions times a character of $\mathrm{Sp}(2)$ at ξ_q , which is the case $Y = C_1$. When λ has at most v parts and equals its own q -core, [2, Cor. 5.5] gives directly $\chi_\lambda(g_{p/q}) = \chi_\lambda^{C_v}(g_{p/q}^{C_v})$, and then $\Delta_\lambda = 1$. For GL_n , the vanishing and the factorisation of $s_\lambda(1, \xi, \dots, \xi^N)$ at a root of unity ξ , with N arbitrary, go back to Littlewood [14, §7.2] (see [4, p. 33]).

12. THE POINTS WHERE THE CLOSED FORM HOLDS

In this section $m \geq 2$, $q \geq 2$, $m = uq + v$ with $0 \leq v < q$, $a = \lfloor (q-1)/2 \rfloor$, and a dominant weight λ is *surviving* if $N_q(\ell) = r_q$. For such λ the proof of Theorem 11.1 gives

$$(8) \quad \chi_\lambda(g_{p/q}) = (-1)^{E_{p,q}(\lambda)} \frac{Z_q(\ell)}{Z_q(\rho)} R_p(b), \quad R_p(b) = \frac{\Pi^Y(b)}{\Pi^Y(\rho_Y)},$$

with $Z_q(\ell)/Z_q(\rho) = \kappa \Delta_\lambda$; so the closed form of Theorem A(2) holds at (λ, p) exactly when $R_p(b) = 1$. Since $\sin(A-B)\sin(A+B) = \frac{1}{2}(\cos 2B - \cos 2A)$, the roots $e_i \pm e_j$ of Y contribute in pairs, and with $c_i = \cos(2\pi p x_i/q)$

$$(9) \quad \Pi^{D_s}(x) = 2^{-\binom{s}{2}} \prod_{i < j} |c_i - c_j|, \quad \Pi^{C_s}(x) = \Pi^{D_s}(x) \prod_i |\sin(2\pi p x_i/q)|.$$

Proof of Theorem C. The four families give the identity. If $q \mid m$, $q \mid 2m+1$ or $q \mid 2m+2$, then $v \in \{0, a, q-1\}$ and $R_p(b) = 1$ by Theorem 11.1. If $q = 6$ and $m \equiv 1 \pmod{6}$, then $v = 1 \leq a = 2$, $Y = C_1$, $\rho_Y = (1)$ and $b = (b_1)$ with $b_1 \in \{1, 2\}$, so by (9) $R_p(b) = |\sin(\pi p b_1/3)|/|\sin(\pi p/3)| = 1$, since $p \equiv \pm 1 \pmod{6}$.

No other point does. Suppose that the identity holds to sign at $p = 1$ for every surviving λ . Then every $\chi_\lambda(g_{1/q})$ is rational, the non-surviving ones being 0 by Proposition 4.1, and an algebraic integer, hence an integer; by Proposition 9.1, q divides $2m$, $2m+1$ or $2m+2$, or $q = 6$ and $m \equiv 1 \pmod{3}$. The cases not listed in Theorem C are (i) $q \geq 4$ even, $q \mid 2m$ and $q \nmid m$, that is $m \equiv q/2 \pmod{q}$, and (ii) $q = 6$ and $m \equiv 4 \pmod{6}$; for $q = 2$ either $q \mid m$ or $q \mid 2m+2$. In both we take the natural representation, $\lambda = (1, 0, \dots, 0)$, and $\xi = e^{2\pi i/q}$. Its character is $\sum_{j=1}^m (\xi^j + \xi^{-j})$, and full turns contribute 0, so it equals $\sum_{j=1}^v (\xi^j + \xi^{-j})$. In case (i), $v = q/2$ and

this is $\sum_{j=1}^{q-1} \xi^j + \xi^{q/2} = -2$; in case (ii), $v = 4$ and it is $2\sum_{j=1}^5 \xi^j - (\xi^5 + \xi^{-5}) = -2 - 1 = -3$. In particular λ is surviving. On the other hand $\ell = \rho + e_1$, the roots not involving the index 1 take the same values on ℓ and on ρ , and those involving it take the values $\{1, \dots, 2m\} \setminus \{m\}$ on ρ and $(\{2, \dots, 2m\} \setminus \{m+1\}) \cup \{2m+2\}$ on ℓ . None of $1, m, m+1, 2m+2$ is divisible by q : in case (i) they are congruent to $1, q/2, q/2+1, 2$, and in case (ii) to $1, 4, 5, 4$. So the two sides have the same multiples of q , $Z_q(\ell) = Z_q(\rho)$, and the identity would give ± 1 . The failure holds at every p , not only at $p = 1$: since $\chi_\lambda(g_{p/q}) = \sigma_p(\chi_\lambda(g_{1/q}))$ for the automorphism σ_p of $\mathbb{Q}(\xi_q)$, a rational value at p forces a rational value at 1, and the natural representation gives -2 , respectively -3 , at every p . $\square$

Proof of Theorem D. By Proposition 9.1, $g_{p/q}$ is conjugate to $g_{1/q}$, so in (8) the value $|\chi_\lambda|$ and the factor $\kappa\Delta_\lambda$ do not depend on p , and $c_\lambda = \kappa R_1(b)$. If $q \mid m$, $q \mid 2m+1$, or $q = 6$ and $m \equiv 1 \pmod{6}$, then $v \leq a$, so $\kappa = 1$, and $R_1(b) = 1$ by the proof of Theorem C. If $q \mid 2m+2$ this is Theorem B. Two cases remain.

$q \geq 4$ even, $q \mid 2m$, $q \nmid m$. Put $h = q/2$. Then $v = h > a$, $s = q - v = h$ and $Y = D_h$. The folded classes are $0, 1, \dots, h$, and exactly one of them, say j , holds its full number of entries; b lists the other h , while $\rho_Y = (h-1, \dots, 0)$ omits $j = h$. With $p = 1$ the numbers $t_k = \cos(\pi k/h)$, $0 \leq k \leq h$, are distinct, and with $\omega(x) = \prod_{k=0}^h (x - t_k)$ the product of $|t_i - t_k|$ over the pairs not containing j is the product over all pairs divided by $|\omega'(t_j)|$. By (9),

$$R_1(b) = \frac{|\omega'(t_h)|}{|\omega'(t_j)|}.$$

The roots of ω are ± 1 and the zeros of the Chebyshev polynomial U_{h-1} , so ω is a constant multiple of $(x^2 - 1)U_{h-1}(x)$, which equals $-\sin \theta \sin h\theta$ at $x = \cos \theta$. At an interior node $\theta = \pi k/h$, $0 < k < h$, its derivative in x has absolute value h ; at $x = \pm 1$ it is $\pm 2U_{h-1}(\pm 1)$, of absolute value $2h$, since $U_{h-1}(\pm 1) = (\pm 1)^{h-1}h$. Hence $R_1(b) = 1$ if $j \in \{0, h\}$ and $R_1(b) = 2$ otherwise. Since $\kappa = 2$ exactly when the class 0 is full, that is $j = 0$, one gets $c_\lambda = 1$ for $j = h$ and $c_\lambda = 2$ for every other j .

$q = 6$, $m \equiv 4 \pmod{6}$. Here $v = 4 > a = 2$, $s = 2$, $Y = D_2$ and $\rho_Y = (1, 0)$; two of the classes $0, 1, 2, 3$ fall one entry short, and $b = (b_1, b_2)$ lists them. With $p = 1$, (9) gives $R_1(b) = 2 |\cos(\pi b_1/3) - \cos(\pi b_2/3)|$, and $\kappa = 2$ exactly when $0 \notin b$:

b	$(1, 0)$	$(2, 0)$	$(3, 0)$	$(2, 1)$	$(3, 1)$	$(3, 2)$
$R_1(b)$	1	3	4	2	3	1
κ	1	1	1	2	2	2
c_λ	1	3	4	4	6	2
λ ($m = 4$)	0	(1)	(1, 1)	(2)	(2, 1)	(2, 2)
$\chi_\lambda(g_{1/6})$	1	-3	4	4	-6	2

The last two rows give, for $m = 4$, a weight realising each column, with $\Delta_\lambda = 1$; the values are computed exactly in $\mathbb{Z}[\zeta_6]$ (`certifica_testigos.py`).

In the last two cases every occupancy pattern of a surviving weight, that is every choice of the classes that fall short, is the pattern of some $\lambda + \rho$, by the argument in the proof of Corollary 5.1; so every listed value of c_λ occurs. $\square$

At $q = 6$ and $m = 4$, for instance, the natural representation has $b = (2, 0)$ and value -3 (proof of Theorem C), and $\Delta_\lambda = 1$, $c_\lambda = 3$.

13. QUESTIONS

Question 13.1. Is there a reductive group H , together with a homomorphism relating it to $\mathrm{Sp}(2m)$, such that the product in Theorem B is the dimension of an irreducible representation of H attached to λ by restriction or by a functor? The example at the end of Section 8 shows that the centraliser of a_P^k does not serve, and the factor SO_{2n+1} on the class $q/2$ points to the dual group, where [18] read their counts. The equality of numbers in Theorem B does not by itself provide such a group, a map or multiplicities.

Question 13.2. Does Theorem A(2) hold outside type C ? The sign part of the proof does not use the type, and the first identity of Remark 6.2 holds in every type; what is missing is the equality $\mathcal{R}_\ell = \mathcal{R}_\rho$ for the surviving weights outside the orbit of ρ under the Weyl group and the translations by q times the coroot lattice.

14. COMPUTATIONS

Every statement above is proved; the computations below were run as independent checks, and the programs with their archived output are included as supplementary files. The criterion (4) was checked on 27 183 dominant weights of $\mathrm{Sp}(2m)$, $m = 2, \dots, 6$, $q = 2, \dots, 14$ (`rho_line_any_q.py`). Theorem A, value and sign, was checked on 18 904 surviving weights of $\mathrm{Sp}(2m)$, $m = 2, \dots, 8$, at every k , against the character computed as an exact integer determinant from the spectrum of Proposition 3.1 (`signo_Fq.py`); the sign form of Remark 7.2 and the identity of Proposition 6.1 on all 11 972 surviving weights with $m = 2, \dots, 7$ (`valor_signo_recuento.py`); Theorem B on the same 11 972 weights (`dims_recuento.py`); Proposition 9.1 on 11 800 pairs (m, q) with $q \leq 60$ and $m \leq 200$, and against the characters themselves on 44 pairs (`galois.py`); Theorem 10.1 against exhaustive enumeration on 40 pairs (m, q) (`valor_signo_recuento.py`). Theorem 11.1, value and sign, was checked in all 4 507 surviving cases (λ, p) with $m = 2, \dots, 5$, $q = 2, \dots, 11$, p prime to q and $\lambda_1 \leq 4$, against characters of $\mathrm{Sp}(2m)$, $\mathrm{Sp}(2s)$ and $\mathrm{SO}(2s)$ evaluated by the Weyl character formula as quotients of determinants at a generic perturbation of the element, with dimensions computed independently; replacing the entries of ρ_{D_s} by $s, \dots, 1$ breaks the formula in 1 224 of these cases (`valor_todo_q.py`). The same program checked it in the 600 cases with $(m, q) = (3, 3), (5, 5), (2, 5), (4, 3), (6, 13), (11, 7)$ and $\lambda_1 \leq 4, 3, 4, 4, 3, 2$ respectively, among them 192 with $u = 1$, $Y = D_3$ and $b \neq \rho_Y$; in the same run Corollary 11.2 was checked on the 312 cases of the first five pairs, and the equality $\chi_\lambda(g_{p/q}) = \chi_\lambda^{C_v}(g_{p/q}^{C_v})$ with $\Delta_\lambda = 1$ for λ equal to its q -core with at most v parts on 164 cases (`valor_todo_q.py casos`). An earlier form of its absolute value, at $p = 1$, was checked on 1 814 surviving weights with $m \leq 6$ and $q \leq 12$ (`residual_factorizacion.py`). Theorem C was checked on the 203 pairs (m, q) with $m = 2, \dots, 8$ and $2 \leq q \leq 30$, on 38 017 surviving cases (λ, p) , with the characters computed exactly in $\mathbb{Z}[X]/(\Phi_q)$, and the natural representation on the 56 pairs of cases (i) and (ii) with $m \leq 30$ (`formula_pura.py`); Theorem D, with the table of its proof, on 52 928 pairs (λ, q) at the points of Proposition 9.1, and Theorem 11.1 once more on 11 244 cases with $m \leq 6$, $q \leq 16$ (`prefactores.py`); both programs use their own evaluator `evlib.py`.

USE OF AI TOOLS

The author used Claude (Anthropic) and ChatGPT (OpenAI) as assistants under his direction: to search the literature, to suggest proof strategies, to write and run the verification programs of

Section 14, and to draft and revise prose. Every proof in the paper was checked by the author, who takes full responsibility for its content.

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